FEATURE PYRAMID NETWORKS

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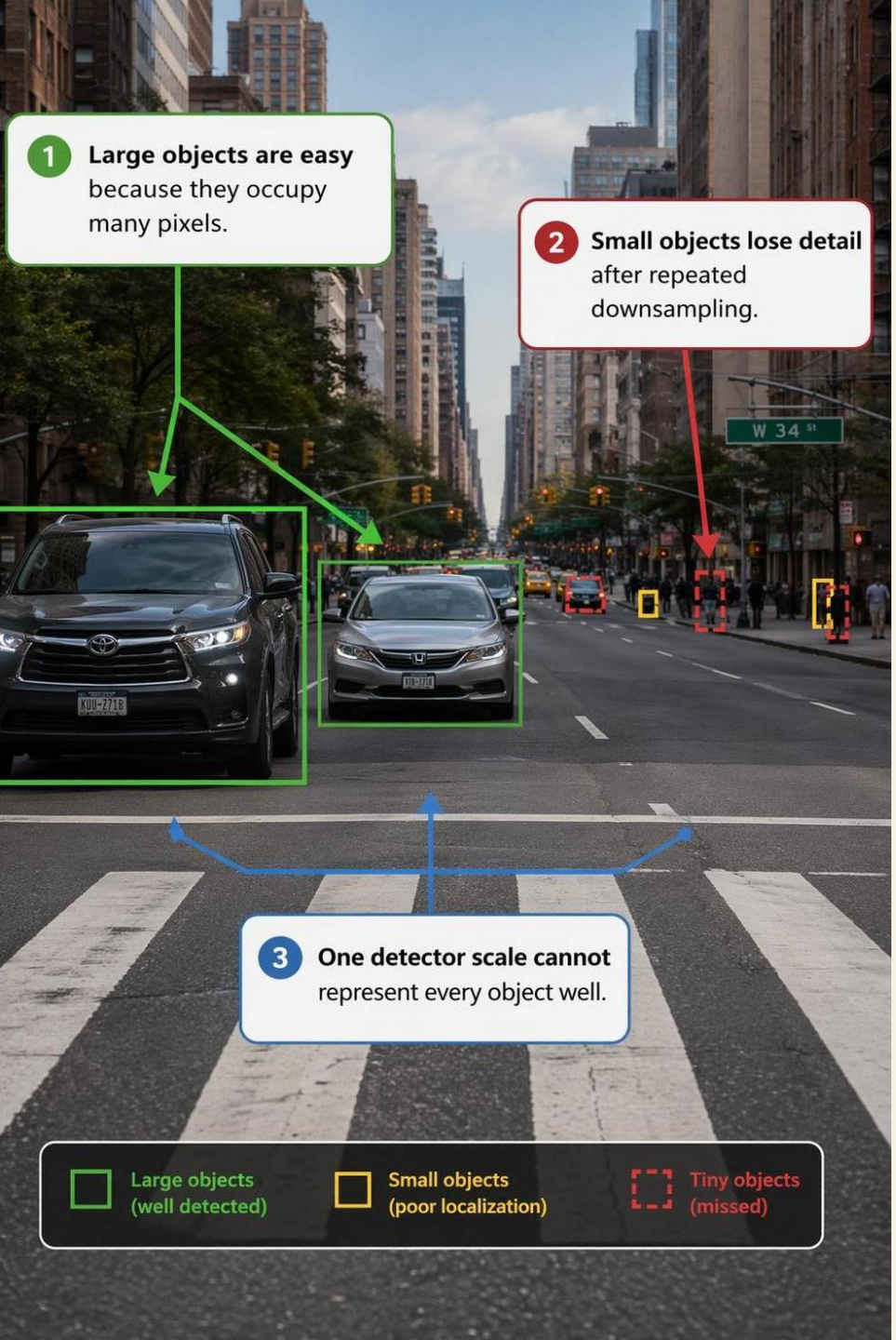
A visual walkthrough of multi-scale detection



WHY SCALE BREAKS DETECTORS

Large objects are easy because they occupy many pixels.

Small objects lose detail after repeated downsampling.

One detector scale cannot represent every object well.

The core problem is not just recognition—it is preserving useful features across scale.

CORE IDEA OF FPN

CNN Backbone

Bottom-up: resolution shrinks, semantics grow; blue feature stages on left

Lateral Connections

Lateral connections: combine meaning with detail at matching stages

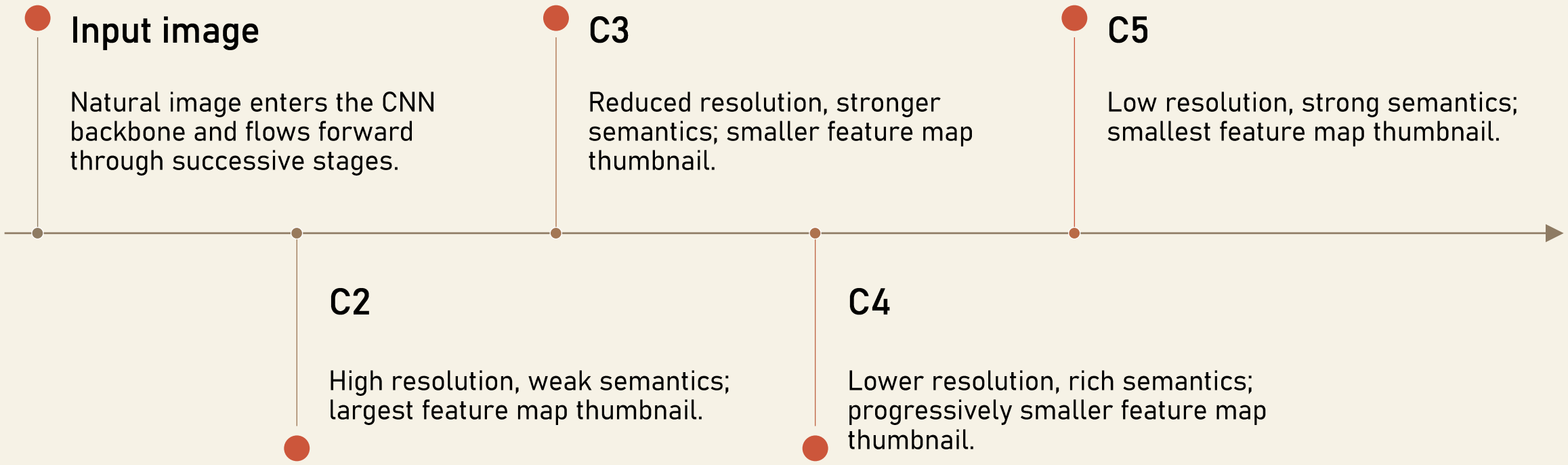
Top-Down Pathway

Top-down: high-level meaning is upsampled; orange flow returns upward

Pyramid Outputs

Green merged outputs on right labeled P2, P3, P4, and P5

STEP 1: BOTTOM-UP PATHWAY



STEP 2: TOP-DOWN PATHWAY

C5 Deep Feature Map

Deep layers know what the object is

×2 Upsample

Pour semantic information back down onto a finer grid

Higher-Resolution Map

Meaning is transferred to shallower spatial locations

×2 Upsample

Upsampling spreads that meaning to finer spatial locations

Top-Down Fusion

Top-down fusion helps smaller objects benefit from stronger semantics.

STEP 3: LATERAL CONNECTIONS

Top-Down Stream

Coarse, semantically strong feature stream runs horizontally across the diagram; Top-down gives meaning.

Lateral Path

Detailed backbone features enter from the side at matching scales; Lateral path gives localization detail.

Scale-Wise Merge

At each scale, show a 1×1 lateral connection feeding a merge node labeled + to create a cleaner combined map.

Before vs After

Use side-by-side mini-panels for one representative level: before merge on the left and after merge on the right.

Detection Output

Merged feature becomes useful for detection.

THE FINAL PYRAMID: P2 TO P5

P2

Small objects

Highest spatial resolution output feature map

Same strong semantics across levels

P3

Medium-small

High-resolution output feature map

Different spatial resolutions for different object sizes

P4

Medium-large

Lower-resolution output feature map

Detectors can choose the right level

P5

Large objects

Lowest-resolution output feature map

Instead of forcing one feature map to do everything

FPN VS. IMAGE PYRAMIDS

1

IMAGE PYRAMIDS

Resize the same image to multiple scales

2

FPN

Show one input feeding a shared CNN backbone

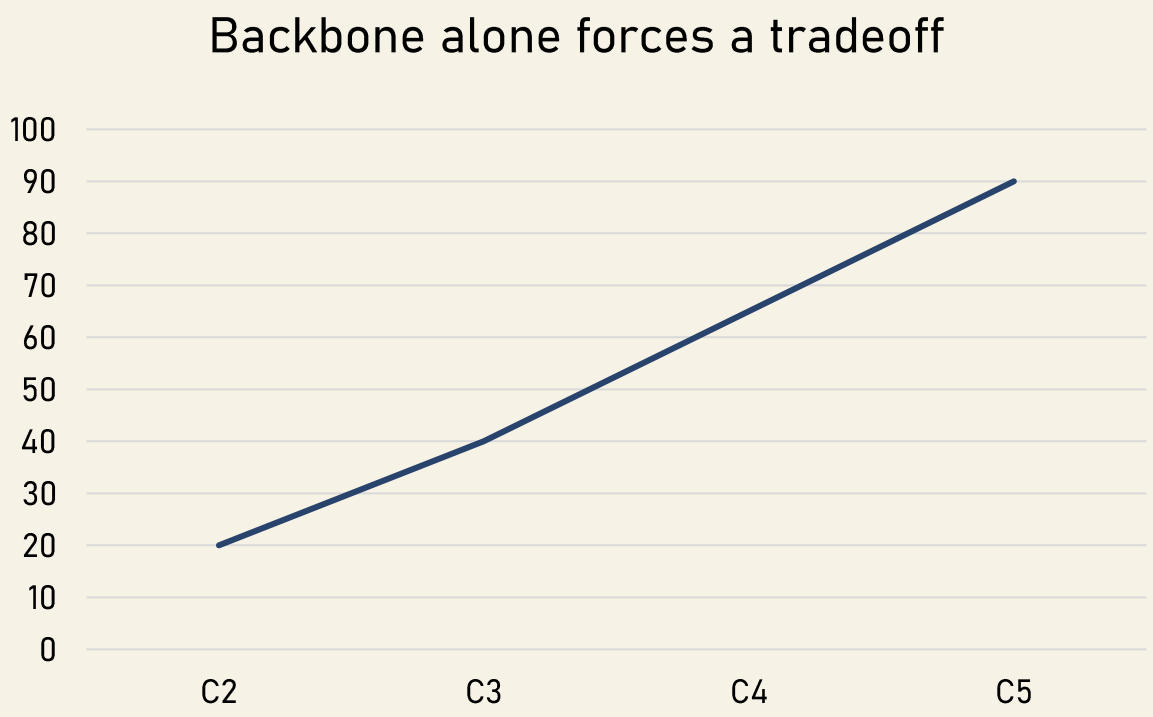
3

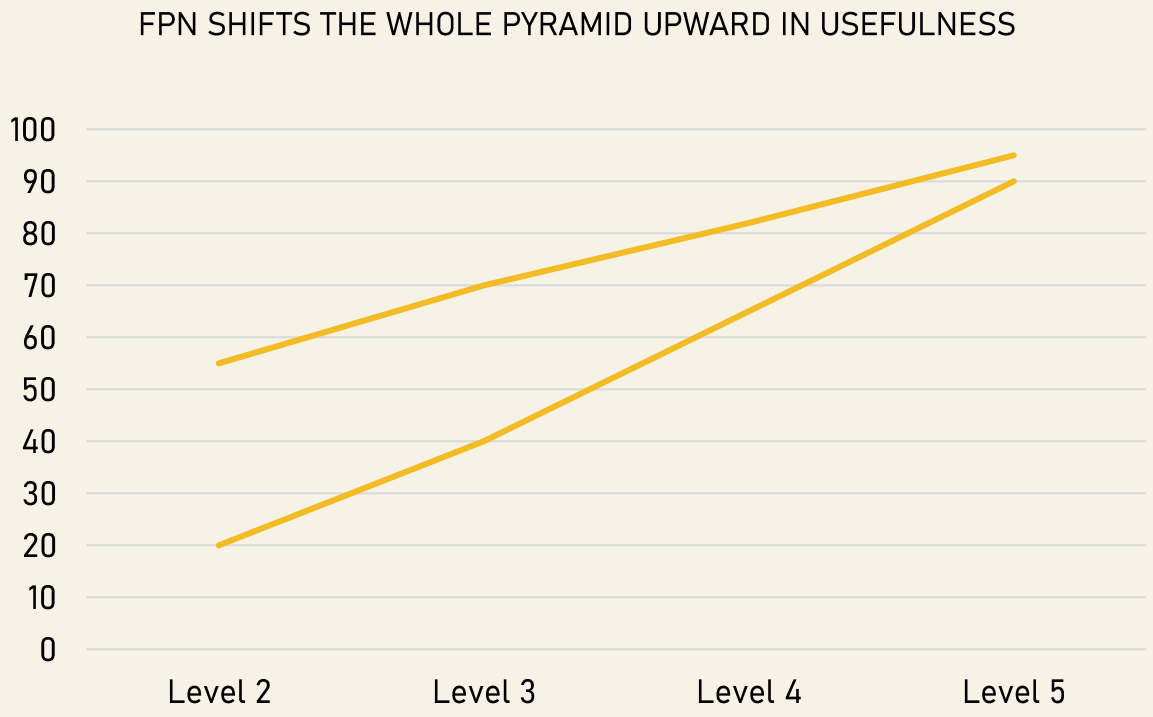
FPN ADVANTAGES

Badge: Speed

Visual comparison of repeated multi-scale backbone passes versus single-pass FPN reuse, with FPN favored on speed, efficiency, and semantic consistency.

THE TRADEOFF FPN FIXES





Left: backbone features trade spatial detail for semantics as depth increases. Right: FPN preserves multi-scale resolution while lifting semantic richness at every level, making the full pyramid more useful for detection.

CASE STUDY: AUTONOMOUS DRIVING

A self-driving system must detect both large nearby vehicles and tiny distant hazards in the same frame. Using the backbone alone makes this difficult: deep features carry strong semantics but become too coarse for small faraway objects, while shallow features preserve detail but lack semantic strength. FPN solves this by combining levels so fine-resolution, semantically enriched maps handle small distant objects and coarser maps handle larger nearby objects more reliably.

In driving, multi-scale awareness is not a convenience—it directly affects safety.

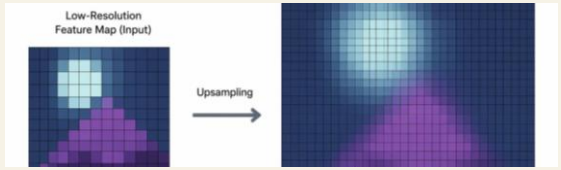
FAILURE CASES AND LIMITATIONS

Static grid assumptions

FPN features still live on discrete grids, so irregular or rotated objects may not align well with the sampled locations.

As a result, localization quality can drop when the object geometry conflicts with standard feature-grid assumptions.

FPN improves scale handling, but it does not remove every representation bottleneck.



Upsampling can blur detail

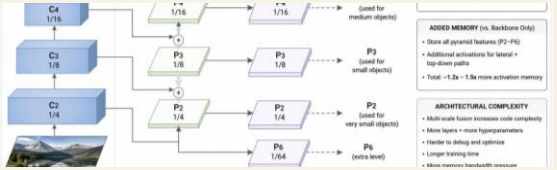
Top-down fusion often enlarges coarse features, which can smooth away sharp edges and fine boundaries.

This helps scale coverage, but it can reduce spatial precision for tasks that depend on exact localization.

Extreme aspect ratios

Very long thin structures are difficult to capture compactly when pyramid levels are tuned for more balanced object shapes.

They may fragment across levels or lose continuity, making representation less stable than for compact objects.



Extra complexity and memory

The added lateral and top-down paths improve multi-scale reasoning, but they also increase feature movement through the network.

That means more compute, more memory pressure, and a heavier deployment footprint than a plain backbone.

WHEN NOT TO USE FPN

Is multi-scale detection the main challenge?

Single-scale sizes: use a simpler detector or plain backbone; FPN may add unnecessary cost.

ALTERNATIVE: WHEN TO MOVE BEYOND FPN

PANet

Adds a second bottom-up path to shorten the flow of low-level localization info to the top layers. Best for precise localization.

BIFPN

Uses weighted bi-directional connections and repetitive stacking for state-of-the-art efficiency. Best for mobile and real time.

GraphFPN

Uses superpixel hierarchies and GNNs to handle non-grid interactions and overlapping objects. Best for irregular clutter.

THANK YOU